

ANEESH RAJ DONTI

LinkedIn | GitHub | Portfolio | aneeshrajdonthi1010@gmail.com | +91 9392576787

WORK EXPERIENCE

AI Engineer— Tata Electronics Pvt. Ltd.

Nov 2025 –present

- Designed and deployed **AI-driven 4D simulation** models using **NVIDIA Omniverse** to **optimize production capacity planning** and operational decision-making at scale.
- Analyzed large point-cloud and operational datasets to perform **planned vs. actual comparisons**, tracking **KPIs (utilization, throughput, cycle time)** and **improving planning accuracy** by **12–18%**.
- Conducted **automated clash detection** and **bottleneck analysis** to identify system inefficiencies early, enabling **15–20%** potential gains in operational efficiency.
- Translated **AI and simulation outputs** into actionable insights through **structured reporting**, **collaborating cross-functionally** with engineering, planning, and operations teams to ensure data accuracy and repeatable workflows.

Business Analytics Intern — Outlook Publishing Group

May 2024– July 2024

- Analyzed and modeled 20,000+ records** using **Python** and **SQL** to **forecast sales trends**, **improving prediction accuracy** by **18%** and supporting data-driven decision-making.
- Performed **exploratory data analysis** and **KPI modeling** to identify demand and engagement drivers, improving campaign efficiency by 15% and customer retention by 12%.
- Designed and automated ETL pipelines** and **analytical workflows**, **reducing manual processing** by 40% and cutting reporting turnaround time from **2 hours** to **30 minutes**.

R&D Intern — NIT Warangal

Nov 2023 – Dec 2023

- Designed and **optimized image processing pipelines** using **Local Binary Patterns (LBP)**, **OpenCV**, and **NumPy**, achieving a 17% runtime improvement through **algorithmic and computational optimizations**.
- Prototyped and evaluated **CNN-based deep learning models** using **TensorFlow**, **PyTorch**, and **Keras**, applying transfer learning to improve image classification performance.
- Curated benchmark datasets (**Corel-250**) and automated model evaluation workflows using **k-NN** and **SVM ranking models**, reducing experiment evaluation time from **2 hours** to **30 minutes** and improving reproducibility.

EDUCATION

National Institute of Technology (NIT), Warangal,
B.Tech in Mechanical Engineering

Dec 2021 – Apr 2025

TECHNICAL SKILLS

Programming & ML: Java, Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL (Joins, Window Functions, Query Optimization)

Cloud & MLOps: AWS (EC2, S3, Lambda, RDS, API Gateway, CloudWatch), Docker, CI/CD Pipelines

Analytics & Visualization: Tableau, Power BI, Excel (Pivot Tables, Power Query, Macros)

Developer Tools: Git, GitHub, Postman, VS Code, Figma

PROJECTS

Real-Time Steel Damage Classification with AWS Cloud Technology | AWS Rekognition, Lambda, S3

- Designed and deployed a **serverless computer vision pipeline** on **AWS** to perform **real-time defect classification**, processing **5,000+ images/day** for industrial quality control.
- Optimized and evaluated **image classification models**, **improving accuracy** from **82%** to **93%** while reducing false negatives in safety-critical inspection workflows.
- Automated **inference and post-processing pipelines** using **parallel execution** and **cloud-native orchestration**, reducing end-to-end analysis time from **2 hours** to **30 minutes** and enabling scalable deployment across multiple production units.

NewsPress – Personalized Content Recommendation System | Python, NLP, Transformers (MiniLM)

- Built a content-based recommender using **MiniLM transformer embeddings** and **cosine similarity** to generate personalized **top-N rankings** on the Microsoft MIND dataset.
- Modeled user preferences from click-stream data and performed **semantic relevance ranking** to improve personalization quality.
- Evaluated **recommendation performance** using **offline ranking** and **click-alignment metrics**, refining embedding and scoring strategies.

Leadership & Organizational Experience

- General Secretary — NSS, NIT Warangal (2023)** – Directed 6+ large-scale community initiatives, including a Blood Donation Camp (350+ units) and Unity Run (500 participants). Spearheaded awareness drives, women's self-defense workshops, and technical talks, engaging 1,200+ students and improving volunteer participation by 35%.
- Head of Logistics — SpringSpree, NIT Warangal (2023)** – Managed logistics operations for South India's 2nd-largest cultural fest (5,000+ attendees), coordinating 100+ volunteers and 20+ vendors. Perfected scheduling and resource allocation, cutting setup time by 2 hours/day and ensuring seamless event execution.